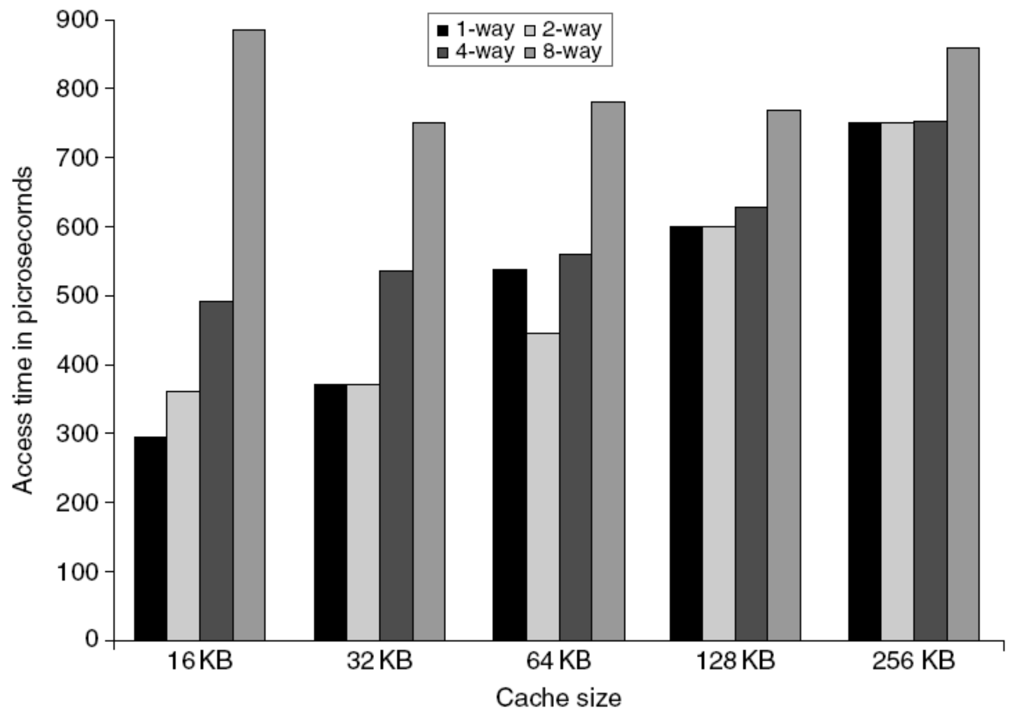
Advanced optimizations of cache performance (§ 2.2)

1. Small and Simple Caches to reduce hit time

- Critical timing path:
  - address tag memory, then compare tags, then select set
- Lower associativity
- Direct-mapped caches can overlap tag comparison and transmission of data



2. Way prediction to reduce hit time

- Combine fast hit time of Direct Mapped and the lower conflict misses of 2-way SA caches?
  - check one way first (speed of direct mapped cache)
  - on a miss, check the other way, if it hits, call it a *pseudo-hit* (slow hit)
  - way prediction is a bit to indicate which half to check first (changes dynamically)

- May extend prediction to more than 2-way SA caches
- Saves power
- Drawback: CPU pipeline is hard if hit takes sometimes 1 and sometimes 2 cycles

3. Pipeline cache access to increase bandwidth

- Examples:
  - » Pentium: 1 cycle
  - » Pentium Pro – Pentium III: 2 cycles
  - » Pentium 4 – Core i7: 4 cycles

A common split is:
• Stage 1: **index the cache and read tags**
• Stage 2: **compare tags** (and pick the right way)
• Stage 3: **read/select data** and forward it

- Increases branch mis-prediction penalty
- Separate the tag compare and data access

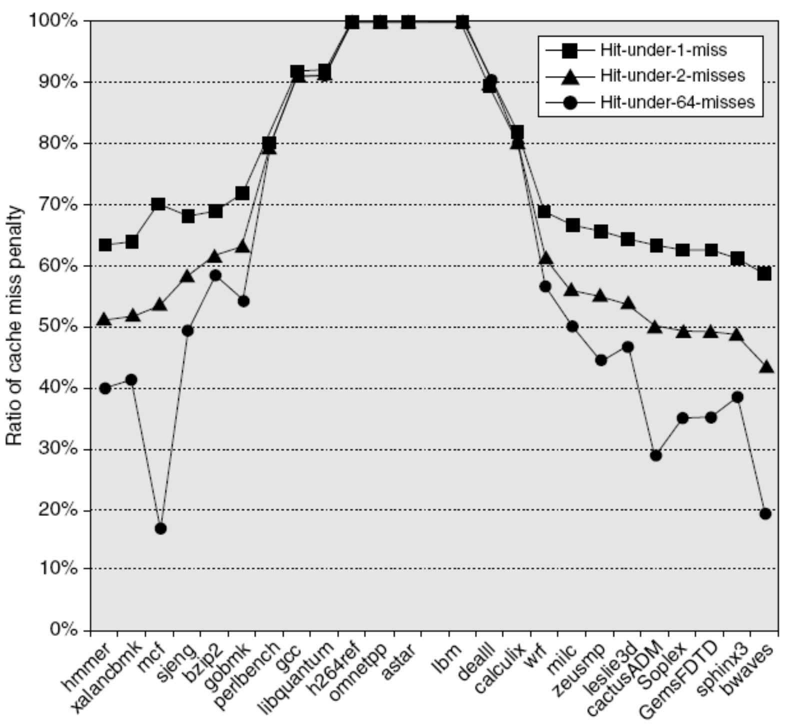
4. Non-blocking caches to increase bandwidth

- *Hit under miss* allows data cache to continue to supply cache hits during a miss -- useful only with out-of-order execution.
- *Hit under multiple miss* or *miss under miss* may further lower the effective miss penalty by overlapping multiple misses

  – Significantly increases the complexity of the cache controller (multiple outstanding memory accesses)

    i.e., MSHR

  – Pentium Pro allows 4 outstanding misses



5. Multi-bank caches to increase bandwidth

- Individual memory controller for each bank.
- Each bank may have its own address and data lines.
- Banks used for independent accesses vs. faster sequential accesses.
  - ARM Cortex-A8 supports 1-4 banks for L2
  - Intel i7 supports 4 banks for L1 and 8 banks for L2
- How blocks are interleaved affects performance.

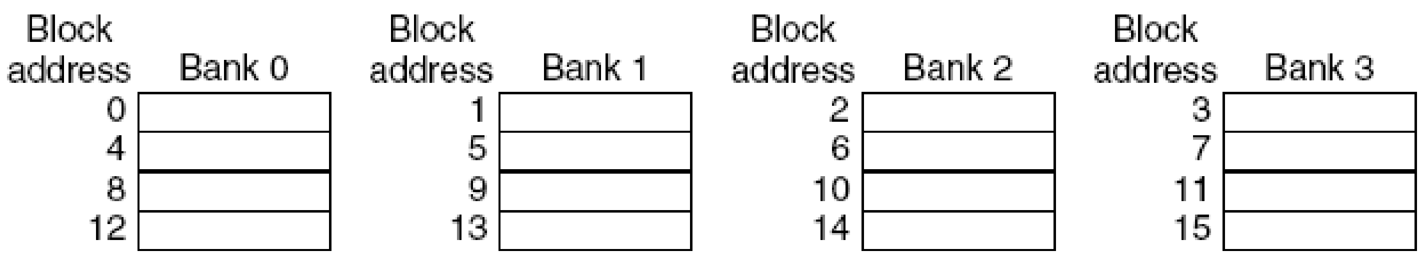


Figure 2.6 Four-way interleaved cache banks using block addressing. Assuming 64 bytes per blocks, each of these addresses would be multiplied by 64 to get byte addressing.

6. Critical word first + early restart to reduce miss penalty

- Don’t wait for full block to be loaded before restarting CPU
  - *Early restart* - As soon as the requested word of the block arrives, send it to the CPU and let the CPU continue execution
  - *Critical Word First* - Request the missed word first from memory and send it to CPU as soon as it arrives; Generally useful only in large blocks,
- Beneficial when we have long cache lines (blocks)
- If want next sequential word, early restart may not be useful,

7. Merging write buffer to reduce miss penalty

No merging

Write address	V		V		V		V	
100	1	Mem[100]	0		0		0	
108	1	Mem[108]	0		0		0	
116	1	Mem[116]	0		0		0	
124	1	Mem[124]	0		0		0	

Merging

Write address	V		V		V		V	
100	1	Mem[100]	1	Mem[108]	1	Mem[116]	1	Mem[124]
	0		0		0		0	
	0		0		0		0	
	0		0		0		0	

- Most useful in write through caches
- Combine writing individual words into a block
- Writing block is faster than writing individual words

8. Compiler optimizations to reduce miss rate

- **Instructions**
  - Reorder procedures in memory so as to reduce conflict misses
  - Aligning basic blocks with cache blocks (lines)
  - Profiling to look at conflicts.

- **Data**
  - *Merging Arrays*: improve spatial locality by single array of compound elements vs. 2 arrays
  - *Loop Fusion*: Combine 2 independent loops that have same looping and some variables overlap
  - *Loop Interchange*: change nesting of loops to access data in order stored in memory
  - *Blocking (Tiling)*: Improve temporal locality by accessing “blocks” of data repeatedly vs. going down whole columns or rows

Merging Arrays Example:

```
int val[SIZE];
int key[SIZE];
```

```
struct merge {
        int val;
        int key;
};
struct merge merged_array[SIZE];
```

Reducing conflicts between val & key improves spatial locality

Loop Fusion Example

```
for (i = 0; i < N; i = i+1)
   for (j = 0; j < N; j = j+1)
         a[i][j] = 1/b[i][j] * c[i][j];
for (i = 0; i < N; i = i+1)
   for (j = 0; j < N; j = j+1)
         d[i][j] = a[i][j] + c[i][j];
```

⇓

```
for (i = 0; i < N; i = i+1)
   for (j = 0; j < N; j = j+1)
   {     a[i][j] = 1/b[i][j] * c[i][j];
         d[i][j] = a[i][j] + c[i][j];}
```

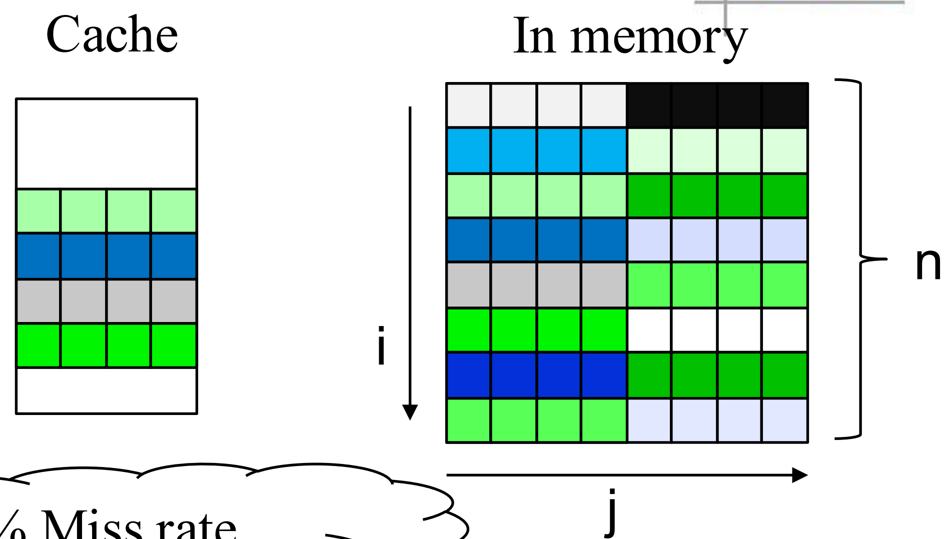
One miss per access to `a` and `c` vs. two misses per access.
Improves temporal locality

loop interchange example

Matrix A is stored Row-wise (row major)

Fully associative cache, block size = 4 words

Cache size < 4n words




```
for (j = 0; j< n; j = j+1)
  for (i = 0; i < n; i = i+1)
       C =+  A[i][j];
```

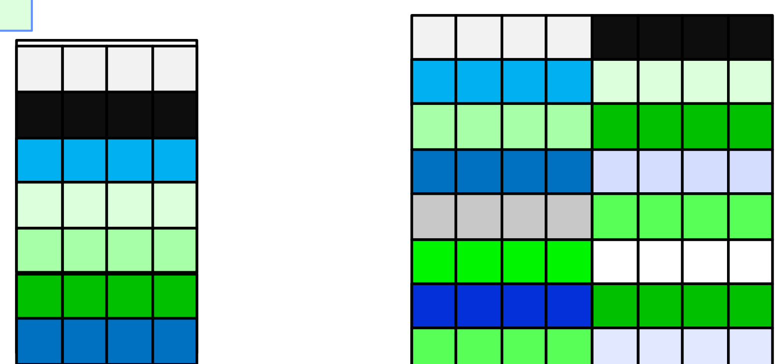
100% Miss rate

Column-wise memory access

Take advantage of spatial locality

Row-wise memory access

```
for (i = 0; i < n; i = i+1)
  for (j = 0; j < n; j = j+1)
       C = + A[i][j];
```

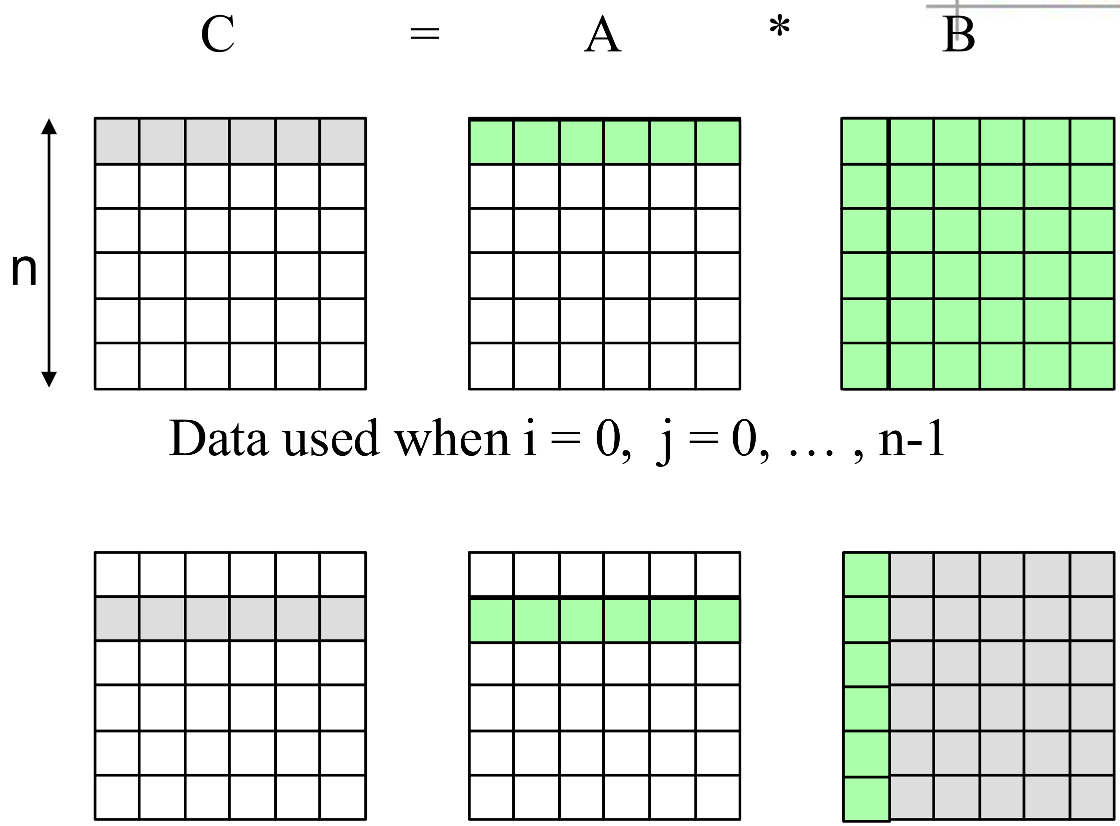


25% Miss rate

Optimization through blocking (partitioning) example

Matrix multiplication

```
for (i = 0 ; i < n ; i++)
  for (j = 0 ; j < n ; j ++)
   {r = 0;
    for (k = 0; k < n; k++)
      r = r + A[i][k]*B[k][j];
   C[i][j] =  r;  };
```



Data used when i = 0, j = 0, … , n-1

Data used when i = 1, j = 0, … , n-1

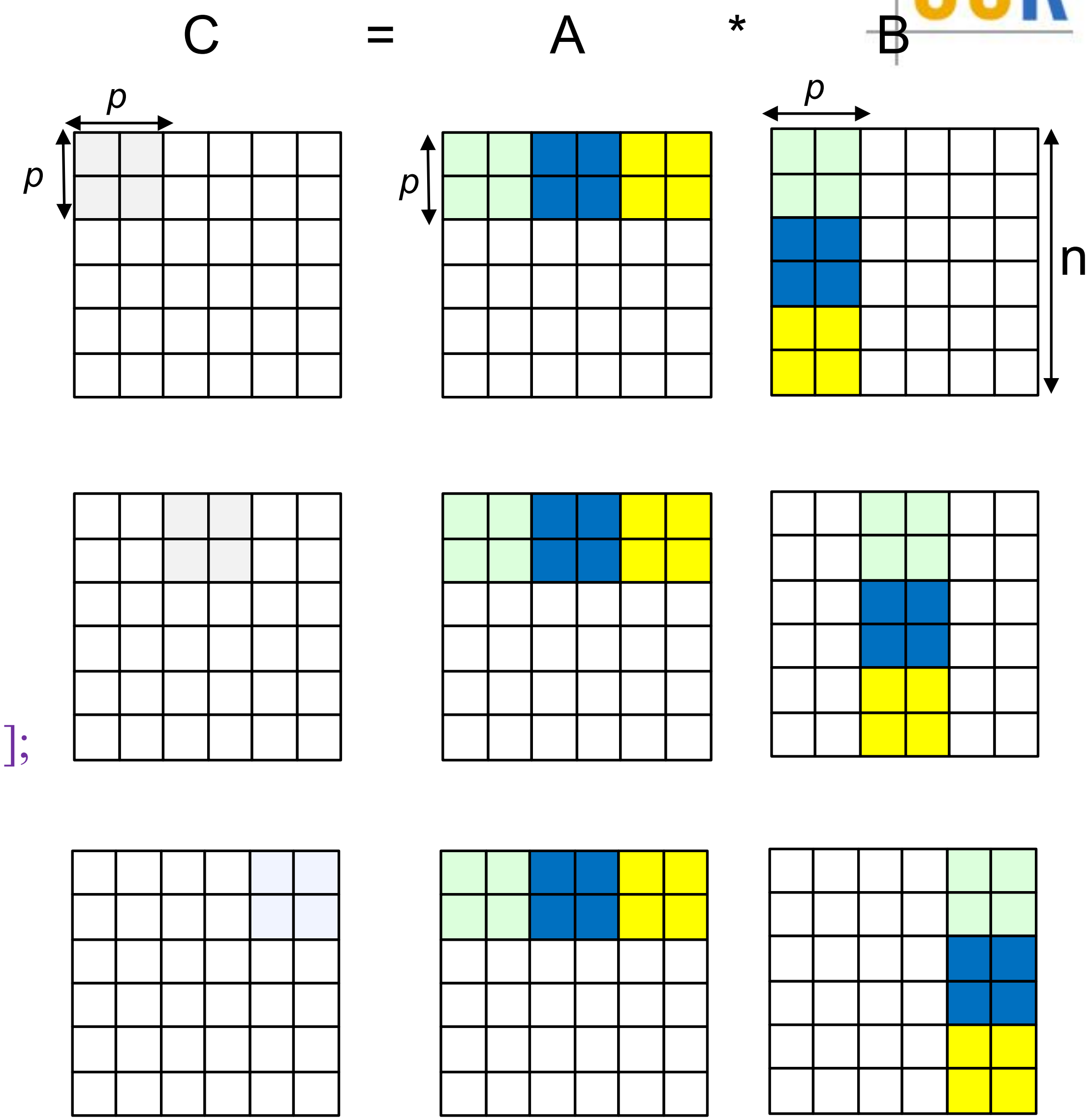
Assume:
A fully associative cache
Block size = 1 word
Cache size < n^2

- One row of A will fit in the cache and be repeatedly used (perfect reuse)
- B will not fit in cache and hence a column of B will be evicted before reuse
- Every element of B will be used only once when brought to the cache

Optimization through blocking (partitioning) example

> Partition the matrices into submatrices of size *p* x *p*

```
for (si = 0; si < n; si =+ p)
 for (sj = 0; sj < n; sj =+ p)
  for (sk = 0; sk < n; sk =+ p)
      for (i=si ; i < si+p ; i++)
        for (j=sj ; j < sj+p ; j++)
          {r = 0;
           for (k=sk; k < sk+p; k++)
                   r = r + A[i][k]*B[k][j];
           C[i][j] =+  r;
          };
```

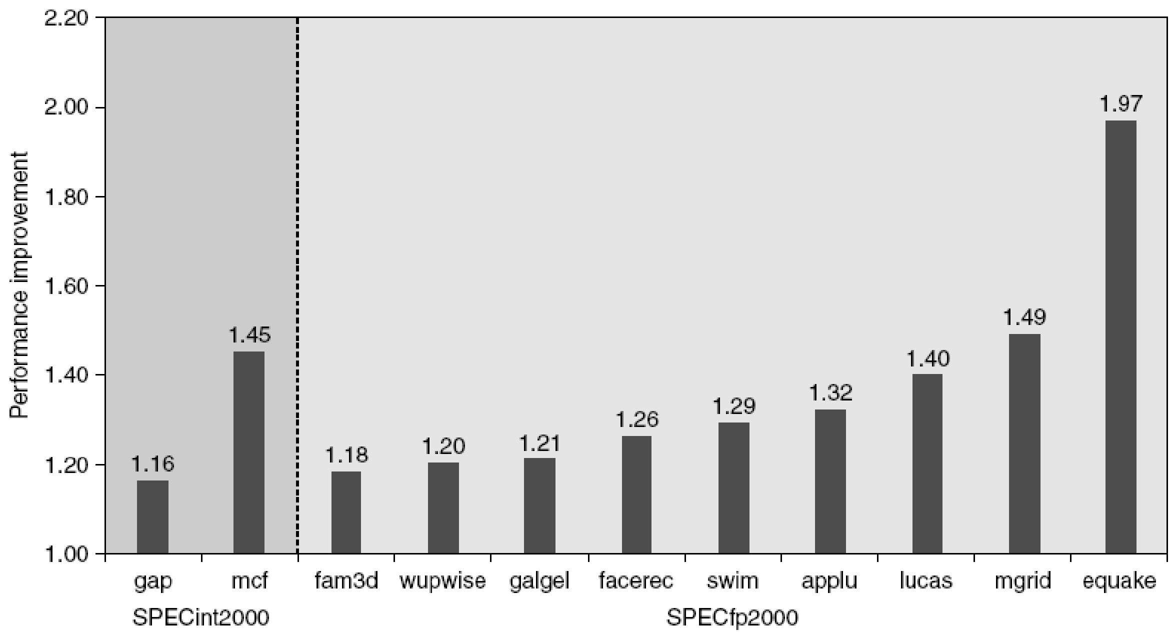


If cache size > $p * n + p^2$, then

- A will be perfectly reused
- Each element of B will be reused "*p*" times (reduce miss rate)

9. Hardware prefetching to reduce miss rate and penalty

- Instruction Prefetching
  - Can fetch 2 (or more) blocks on a miss
  - Extra block placed in "*stream buffer*"
  - On miss, check stream buffer - if found move to cache and prefetch next
- Data Prefetching:
  - May have multiple stream buffers beyond the cache, each prefetching at a different address
- Relies on extra memory bandwidth that can be used without penalty.



10.Compiler prefetching to reduce miss rate and penalty

- Data Prefetch
  - Load data into register
  - Cache Prefetch: load into cache
  - Special prefetching instructions should not cause premature page faults.
  - Issuing Prefetch Instructions takes time
  - Is cost of prefetch issues < savings in reduced misses?
- Works only if can overlap prefetching with execution.

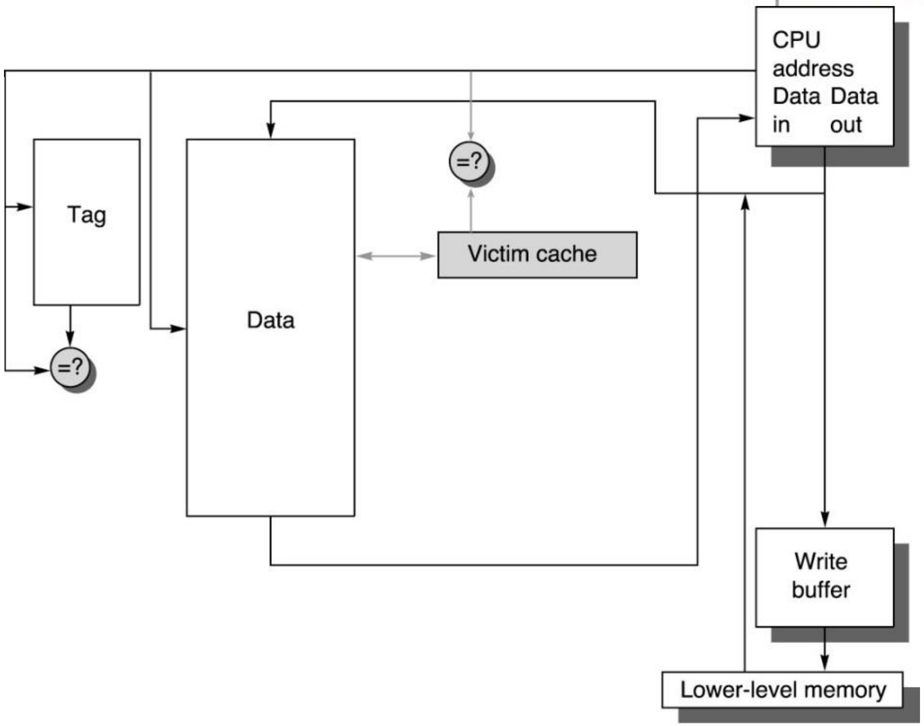
- **Example:** Assume that arrays a[] and b[] are aligned at block boundaries and that the cache block size is 4 words.

```
for (i=0 ; i < 100 ; i++)
    if (i mod 4 = 0) prefetch (b[i+4], a[i+4])
    b[i] = c * a[i] + d * a[i+1] ;
```

if body of loop takes 20 cycles to execute and cache miss penalty is 80 cycles, then, after the first few iterations, data will be in cache when needed.

11. Using victim caches

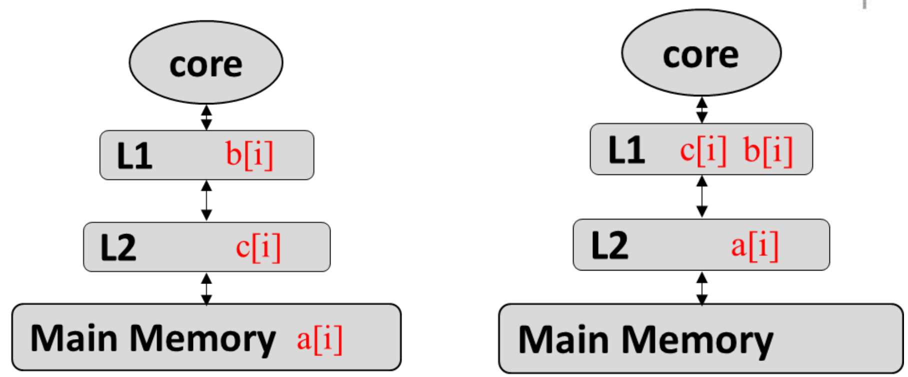
- A buffer to place data just evicted from cache
- A small number of fully associative entries
- Accessed in parallel with cache (no increase in hit time)
- On a hit in the VC, swap blocks in VC and cache



- When used with direct mapped caches, it has the effect of adding associativity to the most recently used cache blocks

12. Computing with Near Data

```
for(i=0; i<n; i++)
    ...
    a[i] = b[i] + c[i] /** Sx */
    ...
    d[i] = a[i] + const /** Sy*/
    ...
endfor
```



- a[i] in Sy can be re-computed by Sx.
- Should we perform the re-computation?
  - Based on the cost evaluation.
  - Only re-computation if the cost is less.

Xulong Tang, Mahmut Taylan Kandemir, Hui Zhao , Myoungsoo Jung, Mustafa Karakoy. “Computing with Near Data.” SIGMETRCIS-2019

13. Other optimizations

- Replacement policy
  - Pseudo LRU
  - RL based
  - Prediction
  - Hot-cold cache
- Multi-directional cache
  - Sumitha George, Minli Julie Liao, Huaipan Jiang, Jagadish B. Kotra, Mahmut T. Kandemir, Jack Sampson, Vijaykrishnan Narayanan: MDACache: Caching for Multi-Dimensional-Access Memories. MICRO 2018: 841-854
  - Minli Julie Liao, Jack Sampson: D-SOAP: Dynamic Spatial Orientation Affinity Prediction for Caching in Multi-Orientation Memory Systems. MICRO 2020: 581-595

Advanced TLB optimizations

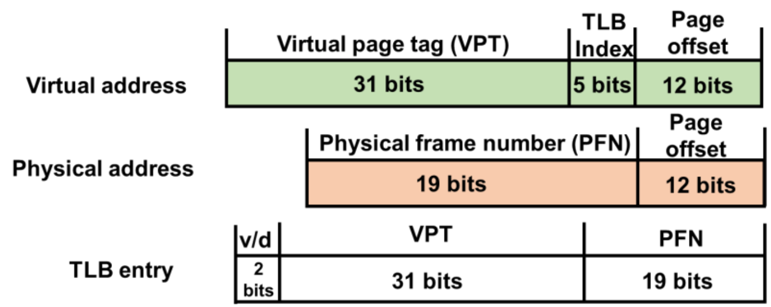
1. Large pages

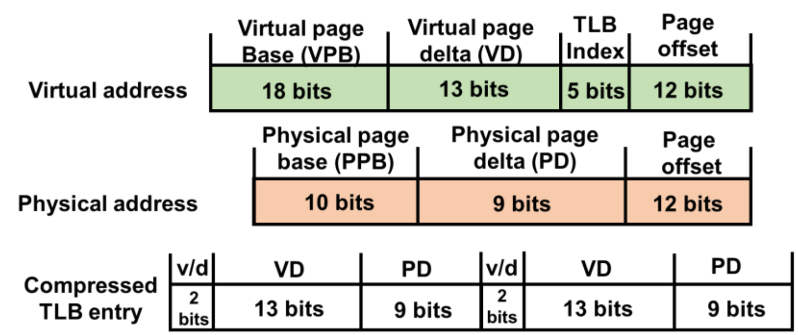
- 4KB vs 2MB vs 1GB
  - + good TLB performance (less memory accesses!)
  - + good for regular access pattern (streaming, sequential)
  - - Internal fragmentation
  - - Expensive data movement and longer accessing time
- Dynamic/multiple page sizes?
  - Rachata Ausavarungnirun, Joshua Landgraf, Vance Miller, Saugata Ghose, Jayneel Gandhi, Christopher J. Rossbach, and Onur Mutlu. 2017. Mosaic: a GPU memory manager with application-transparent support for multiple page sizes. In Proceedings of the 50th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO-50 '17).
  - Guilherme Cox and Abhishek Bhattacharjee. 2017. Efficient Address Translation for Architectures with Multiple Page Sizes. In Proceedings of the Twenty-Second International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS ‘17)
  - Complicated design and software/hardware overheads

2. Compression

- Regular stride page accesses (including consecutive ones)
- Use fewer bits in the VA # and PA #. → more translations in TLB

- Binh Pham, Viswanathan Vaidyanathan, Aamer Jaleel, and Abhishek Bhattacharjee. 2012. CoLT: Coalesced Large-Reach TLBs. In Proceedings of the 2012 45th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO-45)
- B. Pham, A. Bhattacharjee, Y. Eckert, and G. H. Loh. 2014. Increasing TLB reach by exploiting clustering in page translations. In 2014 IEEE 20th International Symposium on High Performance Computer Architecture (HPCA)
- Xulong Tang, Ziyu Zhang, Weizheng Xu, Mahmut Taylan Kandemir, Rami Melhem, and Jun Yang. 2020. Enhancing Address Translations in Throughput Processors via Compression. In Proceedings of the ACM International Conference on Parallel Architectures and Compilation Techniques (PACT '20).





3. Speculative TLB

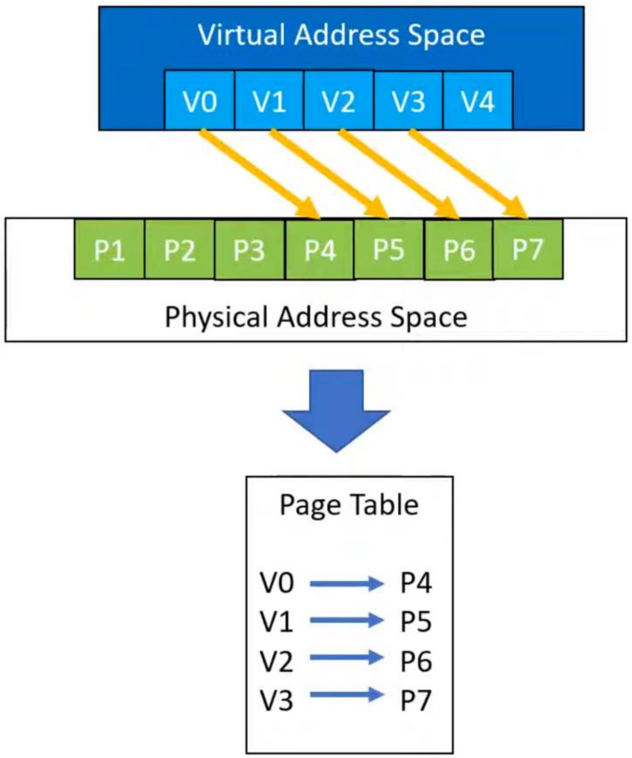
- Similar to cache block prediction and prefetching.
  - Heuristic
  - Machine learning
- Depends on the accuracy and page access pattern.
- T. W. Barr, A. L. Cox, and S. Rixner. 2011. SpecTLB: A mechanism for speculative address translation. In 2011 38th Annual International Symposium on Computer Architecture (ISCA)

4. Range TLB

- Combine consecutive translations to occupy single entry in TLB
- Managed by the OS to determine and memorize the range of both virtual and physical addresses.
- Yan, Zi, et al. "Translation ranger: operating system support for contiguity-aware TLBs." *Proceedings of the 46th International Symposium on Computer Architecture*. (ISCA) 2019.

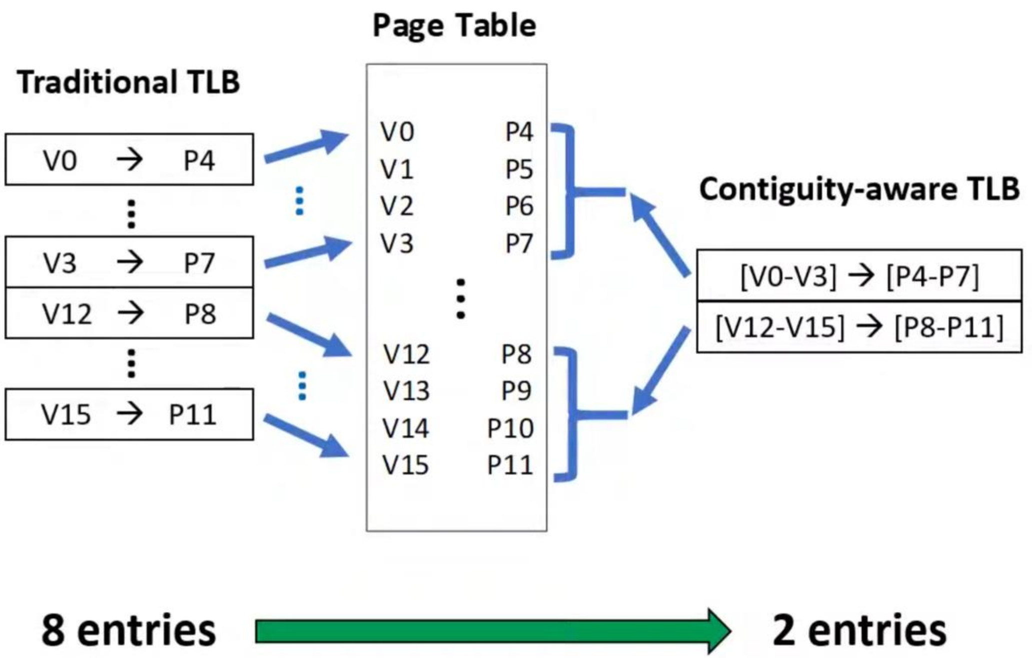
Translation Contiguity

- Address translation contiguity
  - **Contiguous Region**: adjacent virtual pages map to adjacent physical frames
- Can we compress the information?
  - Range-like representation
  - [V0 - V3] -> [P4 - P7]



TLB Coalescing

- Contiguity-aware TLBs cache compressed translations
  - CoLT,
  - MIX TLB,
  - Direct Segment,
  - Devirtualizing Memory,
  - Range TLB, and
  - AMD Ryzen



6. Eager Paging

- Expose range information at allocation, instead of waiting to access (demand paging)
- Allow OS to aggressively optimize the range.
- Vasileios Karakostas, Jayneel Gandhi, Furkan Ayar, Adrián Cristal, Mark D. Hill, Kathryn S. McKinley, Mario Nemirovsky, Michael M. Swift, and Osman Ünsal. 2015. Redundant memory mappings for fast access to large memories. In Proceedings of the 42nd Annual International Symposium on Computer Architecture (ISCA '15)
- Arkaprava Basu, Jayneel Gandhi, Jichuan Chang, Mark D. Hill, and Michael M. Swift. 2013. Efficient Virtual Memory for Big Memory Servers. In Proceedings of the 40th Annual International Symposium on Computer Architecture (Tel-Aviv, Israel) (ISCA '13)
- Swapnil Haria, Mark D. Hill, and Michael M. Swift. 2018. Devirtualizing Memory in Heterogeneous Systems. In Proceedings of the Twenty-Third International Conference on Architectural Support for Programming Languages and Operating Systems (Williamsburg, VA, USA) (ASPLOS '18)